Adaptation of BERT-base-cased for Sentiment Analysis with LoRA and LoRA+

Text-mining & Chatbots (Professor Thomas Gerald)

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1 Introduction

This report presents an adaptation of the BERT-base-cased model for sentiment analysis on the IMDB dataset. Two parameter-efficient fine-tuning methods, LoRA (Low-Rank Adaptation) and LoRA + (Low Rank Adaptation of Large Models), are evaluated and compared against full fine-tuning of BERT-base-cased.

2 Selected Models and Methods

- **Fine Tuned BERT-base-cased**: A transformer-based language model pre-trained on a large corpus, fine-tuned for sentiment classification.
- **LoRA**: Introduces low-rank matrices to reduce trainable parameters that adapt only specific model layers (e.g., attention layers) while maintaining expressiveness and keeping the original model frozen.
- **LoRA+**: An enhanced version of LoRA with further optimizations in memory efficiency and training stability.

3 Experimental Setup

We trained all models using the IMDB dataset from Hugging Face. To ensure a fair comparison, all approaches were trained for **2 epochs** using a batch size of **32** and with a learning rate of **5e-5**. Gradient checkpointing was disabled, and models were evaluated every **128** steps.

The LoRA and LoRA+ hyperparameters were configured as follows:

- **LoRA**:
  - Rank: 8
  - Alpha: 16
  - Dropout: 0.1
- **LoRA+**:
  - Rank: 8
  - Alpha: 32
  - Dropout: 0.1
  - Target Modules: `["query", "value"]`
  - Learning Rate Strategy: Different learning rates for LoRA matrices:
    - $\eta_A = 5e{-}5$ (same as base model)
    - $\eta_B = 24 \times \eta_A = 1.2e{-}3$ (higher learning rate for B)

For full fine-tuning, all model parameters were updated without LoRA adapters. LoRA and LoRA+ only updated the adapter weights, keeping the pre-trained model frozen. All models were trained using the GPU/TPU of Google colab.

4 Results and Discussion

Method	Training Time (s)	Eval Loss ↓	Accuracy ↑	Precision ↑	Recall ↑
Fine-Tuning	2940.13	**0.2199**	**93.60**	**93.73**	**93.46**
LoRA	**2738.39**	0.2520	89.74	89.00	90.70
LoRA+	2739.33	**0.2396**	**90.27**	88.89	**92.05**

Table 1: Comparison of Fine-Tuning, LoRA, and LoRA+

Key Observations

- **Training Speed**: LoRA and LoRA+ train faster (2738s) than full fine-tuning (2940s), reducing training time by approximately 6.8%. LoRA and LoRA+ take nearly the same time, meaning LoRA+ does not add extra computational cost.

- **Evaluation Performance**:
  - Fine-tuning achieves the best accuracy (93.60%), precision (93.73%), and recall (93.46%) but at a higher training cost.
  - LoRA+ outperforms standard LoRA in both evaluation loss (0.2396 vs. 0.2520) and accuracy (90.27% vs. 89.74%).
  - LoRA+ improves recall (92.05%) compared to LoRA (90.70%), indicating better feature learning.
  - LoRA+ slightly reduces precision (88.89% vs. 89.00%), but the difference is minimal.
- **Efficiency**: LoRA+ offers better trade-offs than LoRA—it improves accuracy while maintaining LoRA's speed advantage. Fine-tuning remains the best option if performance is the only concern, but it is more expensive.

Conclusion

- Fine-tuning achieves the best overall performance but requires the longest training time.
- LoRA is faster but sacrifices some accuracy.
- LoRA+ provides a balance—faster training like LoRA but with better accuracy, recall, and lower evaluation loss.
- For a cost-effective alternative to full fine-tuning, **LoRA+ is the better choice over standard LoRA**.